ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

INDUSTRIAL ELECTRONICS N6

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NATED Engineering Studies (Electrical), N6. Operational amplifiers, thyristors and controlled rectifiers, power supplies and regulators, oscillators and timers, transducers. A formula sheet is attached.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations. Write units with all answers.
5. Sketches and circuit diagrams must be neat, large and fully labelled.
6. A formula sheet is attached at the end of this question paper.
7. Round off answers to TWO decimal places, unless stated otherwise.
8. Write neatly and legibly.

QUESTION 1: OPERATIONAL AMPLIFIERS

1.1 An inverting summing amplifier has a feedback resistor of 100 kΩ and three inputs: V_1 = 0,5 V through 20 kΩ, V_2 = −1 V through 50 kΩ, and V_3 = 0,2 V through 25 kΩ. Draw the circuit and calculate the output voltage. (5)

1.2 An op-amp integrator has R = 10 kΩ and C = 0,1 µF and is supplied from ±15 V. A constant input of +2 V is applied at t = 0 with the capacitor initially uncharged.

1.2.1 Calculate the rate of change of the output voltage. (3)

1.2.2 Calculate the output voltage after 2 ms. (2)

1.3 Sketch the output waveform of an op-amp differentiator when a triangular wave is applied to the input. Show the input and output waveforms on the same time axis. (4)

1.4 A non-inverting amplifier has R_f = 90 kΩ and R_i = 10 kΩ and is supplied from ±15 V (saturation at about ±13,5 V).

1.4.1 Calculate the voltage gain. (2)

1.4.2 Calculate the output voltage for an input of 0,4 V. (2)

1.4.3 What will the output voltage be for an input of 2 V? Explain. (2)

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QUESTION 2: THYRISTORS AND CONTROLLED RECTIFIERS

2.1 Sketch the forward and reverse voltage–current characteristic of a silicon-controlled rectifier (SCR). Label the forward breakover voltage, holding current and reverse breakdown region. (5)

2.2 A single-phase half-wave controlled rectifier supplies a 20 Ω resistive load from a 230 V, 50 Hz supply. The firing angle is 60°.

2.2.1 Calculate the average load voltage. (4)

2.2.2 Calculate the average load current. (2)

2.3 A single-phase fully controlled bridge supplies a highly inductive load (continuous current) from the same 230 V supply. Calculate the average output voltage at a firing angle of 30°. (4)

2.4 Explain the difference between an SCR and a triac, and name ONE application of a triac with a diac trigger. (5)

[20]

QUESTION 3: POWER SUPPLIES AND REGULATORS

3.1 A bridge rectifier with a capacitor filter is supplied from a 12 V (RMS), 50 Hz transformer secondary. Each silicon diode drops 0,7 V. The load current is 0,5 A and the filter capacitor is 2 200 µF.

3.1.1 Draw the circuit diagram. (3)

3.1.2 Calculate the peak output voltage. (2)

3.1.3 Calculate the peak-to-peak ripple voltage. (2)

3.1.4 Calculate the approximate average DC output voltage. (1)

3.2 A Zener diode regulator must supply 9,1 V to a 200 Ω load from a 15 V DC supply. The Zener diode needs a minimum current of 5 mA.

3.2.1 Calculate the load current. (2)

3.2.2 Calculate the value of the series resistor. (3)

3.2.3 Calculate the power dissipated in the Zener diode if the load is disconnected. (3)

3.3 Give TWO advantages and TWO disadvantages of a switch-mode power supply compared with a linear power supply. (4)

[20]

QUESTION 4: OSCILLATORS AND TIMERS

4.1 A Wien-bridge oscillator uses R = 10 kΩ and C = 10 nF in its frequency-determining network.

4.1.1 Calculate the frequency of oscillation. (3)

4.1.2 State the voltage gain required from the amplifier, and the ratio of R_f to R_1 needed in a non-inverting configuration. (3)

4.2 A 555 timer is connected as an astable multivibrator with R_A = 1 kΩ, R_B = 10 kΩ and C = 10 µF. Calculate:

4.2.1 the time high and the time low (4)

4.2.2 the frequency and the duty cycle (4)

4.3 State the Barkhausen criteria for sustained oscillation. (2)

4.4 Give TWO reasons why crystal oscillators are used in microcontrollers and clocks. (2)

4.5 Name TWO industrial applications of a 555 timer. (2)

[20]

QUESTION 5: TRANSDUCERS

5.1 A strain gauge with a gauge factor of 2 and a resistance of 120 Ω is bonded to a steel beam. The strain is 500×10^{-6}.

5.1.1 Calculate the change in resistance. (4)

5.1.2 The gauge is one arm of a Wheatstone bridge (quarter bridge) supplied with 10 V. Calculate the approximate output voltage of the bridge. (4)

5.2 Compare a thermocouple, a resistance temperature detector (RTD) and a thermistor under the headings: principle of operation and ONE advantage. (6)

5.3 Explain, with the aid of a labelled sketch, the operation of a linear variable differential transformer (LVDT). (3)

5.4 Explain the difference between an active and a passive transducer and give ONE example of each. (3)

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TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Inverting: $A_v = -\frac{R_f}{R_i}$; non-inverting: $A_v = 1 + \frac{R_f}{R_i}$

Summing: $V_o = -R_f(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3})$

Integrator: $\frac{\Delta V_o}{\Delta t} = -\frac{V_{in}}{RC}$

Differentiator: $V_o = -RC\ \frac{dV_{in}}{dt}$

$V_m = \sqrt{2}\ V_{RMS}$

Half-wave controlled (R load): $V_{av} = \frac{V_m}{2\pi}(1 + \cos \alpha)$

Full-wave fully controlled (inductive load): $V_{av} = \frac{2V_m}{\pi}$ cos α

Full-wave ripple: $V_{r(p\text{-}p)} = \frac{I}{f_r C}$, $f_r = 2f$

$V_{DC} \approx V_p - \frac{V_{r(p\text{-}p)}}{2}$

Zener: $R_s = \frac{V_s - V_z}{I_L + I_{z(min)}}$; $P_z = V_z I_z$

Wien bridge: $f = \frac{1}{2\pi RC}$

555 astable: $t_H = 0{,}693(R_A + R_B)C$; $t_L = 0{,}693R_BC$

Duty cycle = $\frac{t_H}{t_H + t_L} \times 100\%$; $f = \frac{1}{T}$

Strain gauge: GF = $\frac{\Delta R/R}{\varepsilon}$

Quarter bridge: $V_o \approx \frac{V \cdot GF \cdot \varepsilon}{4}$